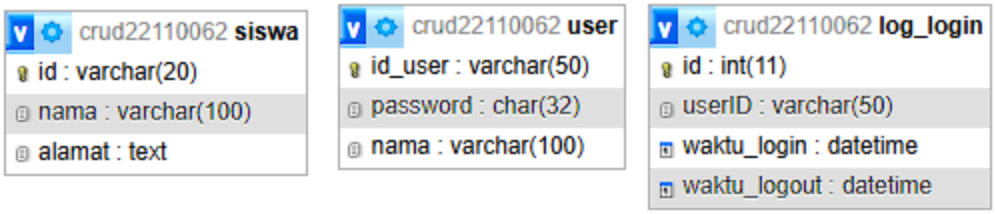
Nama : Pahni Hanawan
Nim : 22110062
Kelas : L 2022
MK : PBO

Modul 13

Database



Session.java

```java
package crudpbo;

public class Session {
    private static String UserID, Nama, StatusLogin;

    public static void setUserID(String UserID) {
        Session.UserID = UserID;
    }

    public static String getUserID() {
        return UserID;
    }

    public static void setNama(String Nama) {
        Session.Nama = Nama;
    }

    public static String getNama() {
        return Nama;
    }

    public static void setStatusLogin(String StatusLogin) {
        Session.StatusLogin = StatusLogin;
    }

    public static String getStatusLogin() {
        return StatusLogin;
    }
}

```

Login.java

```java
package crudpbo;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class Login extends UserBase implements DatabaseAccess {
    private Connection koneksi;
    private PreparedStatement psmt;
    private ResultSet dataUser;
    private String query, pesan;

    public Login() {
        KoneksiMysql koneksi = new KoneksiMysql();
        try {
            this.koneksi = koneksi.getKoneksi();
        } catch (SQLException ex) {
            ex.printStackTrace();
        }
    }

    @Override
    public String cekLogin(String userID, String password) {
        query = "SELECT * FROM user WHERE id_user=? AND password=md5(?)";
        try {
            psmt = koneksi.prepareStatement(query);
            psmt.setString(1, userID);
            psmt.setString(2, password);
            dataUser = psmt.executeQuery();

            if (dataUser.next()) {
                Session.setUserID(userID);
                Session.setNama(dataUser.getString("nama"));
                Session.setStatusLogin("AKTIF");

                // log login
                query = "INSERT INTO log_login (userID, waktu_login) VALUES (?, CURRENT_TIMESTAMP())";
                try {
                    psmt = koneksi.prepareStatement(query);
                    psmt.setString(1, userID);
                    psmt.executeUpdate();
                    psmt.close();
                } catch (Exception e) {
                    pesan = "Gagal Simpan Log Login";
                }
                pesan = "Login berhasil";
            } else {
                pesan = "Gagal Login";
            }
        } catch (Exception e) {
            pesan = "Gagal Login, Query Error";
        }
        return pesan;
    }

    @Override
    public void Logout(String userID) {
        query = "UPDATE log_login SET waktu_logout=CURRENT_TIMESTAMP() WHERE userID=? ORDER BY id DESC LIMIT 1";
        try {
            psmt = koneksi.prepareStatement(query);
            psmt.setString(1, userID);
```

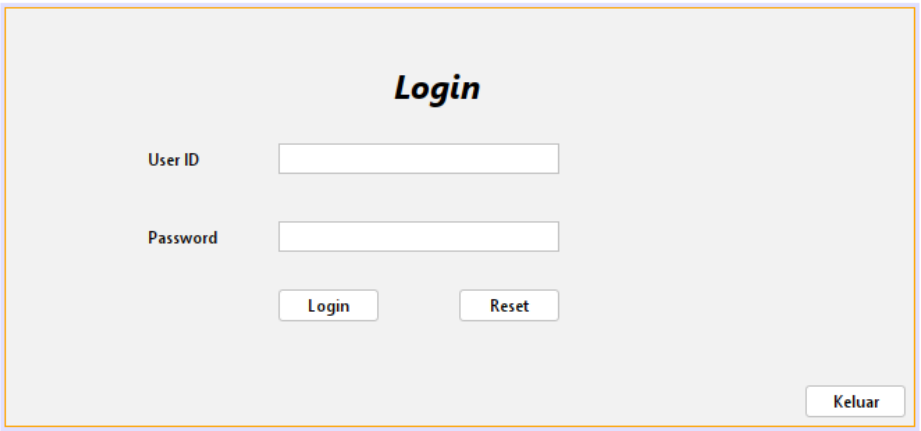
```
63                  psmt.executeUpdate();
64                  psmt.close();
65
66                  // hapus Session
67                  Session.setUserID(null);
68                  Session.setNama(null);
69                  Session.setStatusLogin(null);
70
71                  koneksi.close();
72              } catch (Exception e) {
73                  e.printStackTrace();
74              }
75          }
76      }
77
```

Abstract Class

```
1   package crudpbo;
2
3   public abstract class UserBase {
4       protected String userID;
5       protected String password;
6
7       public String getUserID() {
8           return userID;
9       }
10
11      public void setUserID(String userID) {
12          this.userID = userID;
13      }
14
15      public String getPassword() {
16          return password;
17      }
18
19      public void setPassword(String password) {
20          this.password = password;
21      }
22  }
```

Interaface

```
1   package crudpbo;
2
3   public interface DatabaseAccess {
4       public String cekLogin(String userID, String password);
5       public void Logout(String userID);
6   }
7
```




```
package crudpbo;

import javax.swing.JOptionPane;
import javax.swing.JFrame;

public class LoginForm extends javax.swing.JFrame {
private String pesan;
    // membuat objek baru dari class Login yang sudah terkoneksi dengan database
    Login bb = new Login();

    public LoginForm() {
        initComponents();
    }

    // method untuk mereset text
    public void reset_text() {
        txt_userid.setText("");
        txt_password.setText("");
    }
    @SuppressWarnings("unchecked")
    Generated Code

    private void btn_loginActionPerformed(java.awt.event.ActionEvent evt) {
          if (txt_userid.getText().trim().equals("")) {
            JOptionPane.showMessageDialog(null, "Maaf, User ID belum diisi !");
            txt_userid.requestFocus();
        } else if (txt_password.getText().trim().equals("")) {
            JOptionPane.showMessageDialog(null, "Maaf, Password belum diisi !");
            txt_password.requestFocus();
        } else {
            bb.setUserID(txt_userid.getText());
            bb.setPassword(txt_password.getText());

            pesan = bb.cekLogin(bb.getUserID(), bb.getPassword());

            if (Session.getStatusLogin().equals("AKTIF")) {
                // menutup form login tanpa keluar dari aplikasi
                dispose();

                // memanggil form menu
                Menu form = new Menu();
                form.setVisible(true);
            } else {
                JOptionPane.showMessageDialog(null, pesan, "Informasi", JOptionPane.INFORMATION_MESSAGE);
            }
        }
    }

    private void btn_resetActionPerformed(java.awt.event.ActionEvent evt) {
        reset_text();
    }

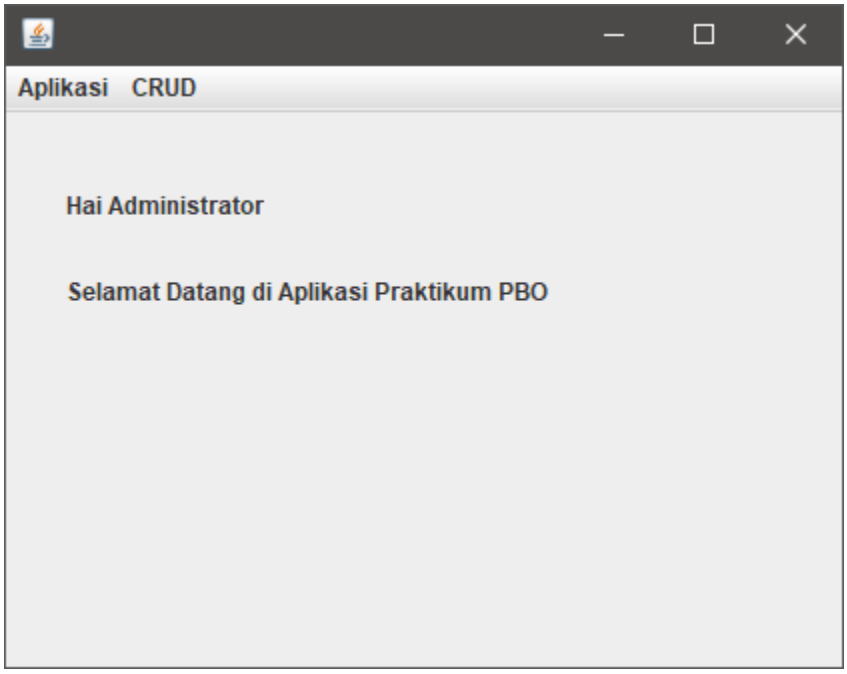
    private void btn_keluarActionPerformed(java.awt.event.ActionEvent evt) {
        if (JOptionPane.showConfirmDialog(null, "Apakah Anda yakin akan keluar ?", "Warning", 2)
            == JOptionPane.YES_OPTION) {
            System.exit(0);
        }
    }

```

```
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        Look and feel setting code (optional)

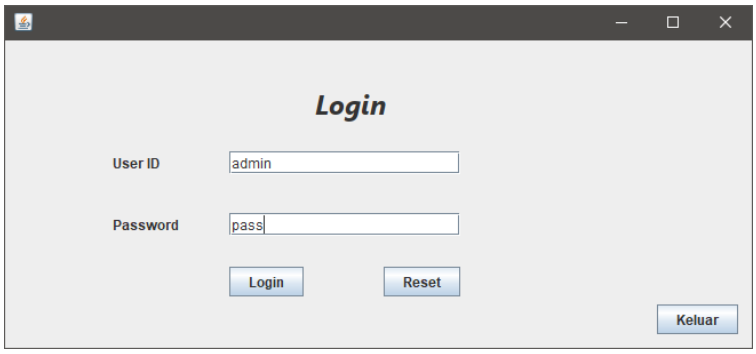
        /* Create and display the form */
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new LoginForm().setVisible(true);
            }
        });
    }

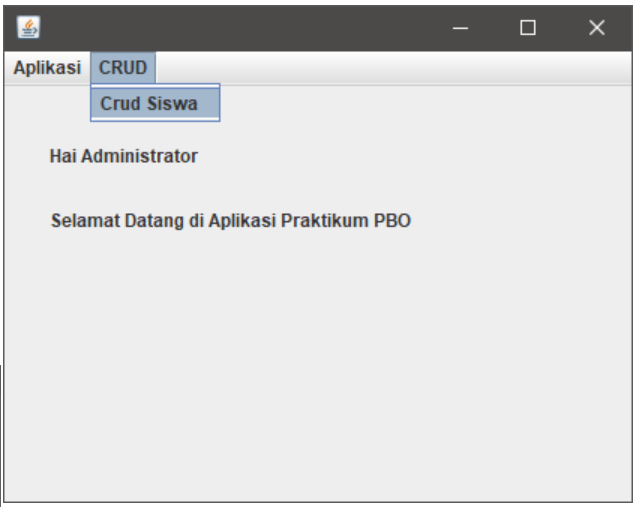
    // Variables declaration - do not modify
    private javax.swing.JButton btn_keluar;
    private javax.swing.JButton btn_login;
    private javax.swing.JButton btn_reset;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JLabel jLabel3;
    private javax.swing.JPanel jPanel1;
    private javax.swing.JTextField txt_password;
    private javax.swing.JTextField txt_userid;
    // End of variables declaration
}
```

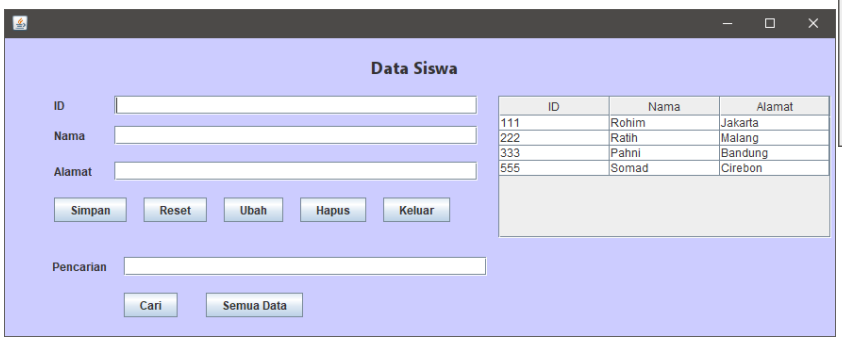


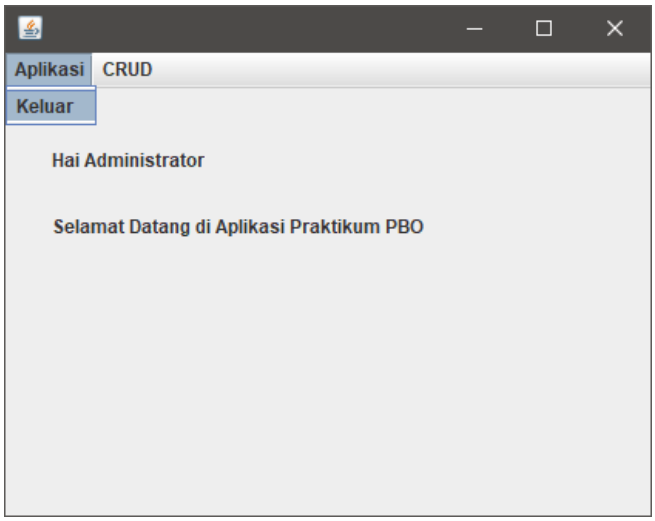
```
1
2   package crudpbo;
3
4   import javax.swing.*;
5
6   public class Menu extends javax.swing.JFrame {
7    // Membuat objek baru dari class Login yang sudah terkoneksi dengan database
8       Login bb = new Login();
9       public Menu() {
10          // Mengecek apakah sudah login atau belum
11          if (Session.getStatusLogin().equals("AKTIF")) {
12              initComponents();
13              setSessionNama();
14          } else {
15              // Menutup menu tanpa keluar dari aplikasi
16              dispose();
17              // Memanggil form login
18              LoginForm form = new LoginForm();
19              form.setVisible(true);
20          }
21      }
22
23      private void setSessionNama() {
24          label_nama.setText("Hai " + Session.getNama());
25      }
26      @SuppressWarnings("unchecked")
27      Generated Code
99
100     private void jMenuKeluarActionPerformed(java.awt.event.ActionEvent evt) {
101       if (JOptionPane.showConfirmDialog(null, "Apakah Anda yakin akan keluar ?", "Warning", JOptionPane.YES_NO_OPTION)
102           == JOptionPane.YES_OPTION) {
103           bb.Logout(Session.getUserID());
104           System.exit(0);
105       }
106   }
107
108   private void jMenuCRUDSiswaaActionPerformed(java.awt.event.ActionEvent evt) {
109       try {
110           Form_Siswa form = new Form_Siswa();
111           form.setVisible(true);
112       } catch (Exception ex) {
113           JOptionPane.showMessageDialog(this, "Gagal membuka Form Siswa: " + ex.getMessage());
114       }
115   }
116
117   /**
118    * @param args the command line arguments
119    */
120   public static void main(String args[]) {
121       /* Set the Nimbus look and feel */
122       Look and feel setting code (optional)
143
144       /* Create and display the form */
145       java.awt.EventQueue.invokeLater(new Runnable() {
146           public void run() {
147               new Menu().setVisible(true);
148           }
149       });
150   }
```

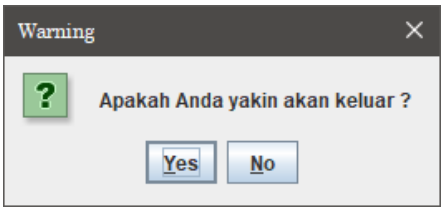
```
151
152         // Variables declaration - do not modify
153         private javax.swing.JMenu jMenuAplikasi;
154         private javax.swing.JMenuBar jMenuBar1;
155         private javax.swing.JMenu jMenuCRUD;
156         private javax.swing.JMenuItem jMenuCRUDSiswaa;
157         private javax.swing.JMenuItem jMenuItem1;
158         private javax.swing.JMenuItem jMenuItem2;
159         private javax.swing.JMenuItem jMenuKeluar;
160         private javax.swing.JLabel label_keterangan;
161         private javax.swing.JLabel label_nama;
162         // End of variables declaration
163     }
164
```


Login
User ID
admin
Password
pass
Login
Reset
Keluar


Aplikasi
CRUD
Crud Siswa
Hai Administrator
Selamat Datang di Aplikasi Praktikum PBO


Data Siswa
ID
Nama
Alamat
Simpan
Reset
Ubah
Hapus
Keluar
Pencarian
Cari
Semua Data
ID
Nama
Alamat
111
Rohim
Jakarta
222
Ratih
Malang
333
Pahni
Bandung
555
Somad
Cirebon


Aplikasi
CRUD
Keluar
Hai Administrator
Selamat Datang di Aplikasi Praktikum PBO


Warning
Apakah Anda yakin akan keluar ?
Yes
No